15

SIS DESIGN CHECKLIST

Introduction

The use of a checklist will not, in and of itself, lead to safer systems, just as performing a HAZard and OPerability study (HAZOP) and not following any of the recommendation will not lead to a safer facility. Following the procedures outlined in the checklist, which are based on industry standards and accumulated knowledge (much of which was learned "the hard way"), should result in safer systems. The checklist is an attempt to list as many procedures and common practices as possible in the hope that, by following a systematic review of the overall design process, nothing will fall through the cracks of an organization and be forgotten.

This checklist is composed of various sections, each corresponding to different portions of the safety lifecycle as described in various standards. Different sections of the checklist are intended for different groups involved with the overall system design, ranging from the user, contractor, vendor, and system integrator. Exactly who has what responsibility may vary from project to project. The checklist, therefore, does not dictate who has what responsibilities, it only summarizes items in the various lifecycle steps.

Why bother? The English Health and Safety Executive reviewed and published the findings of 34 accidents that were the direct result of control and safety system failures. Their findings are summarized in Figure 15-1. Forty-four percent of the accidents were due to *incorrect and incomplete specifications* (functional and integrity), and 20% due to changes made after commissioning. One can easily see that the majority of these issues focus on *user* activities. (Hindsight is always 20/20, foresight is a bit more difficult.) Industry standards, as well as this checklist, attempt to cover *all* of the issues, and not just focus on any one particular area.

This checklist should in no way be considered final or complete. As others review and use it, they are encouraged to add to it (for the benefit of others who may use it in the future). In fact, each section starts with a new page, leaving ample space for your additions and suggestions.

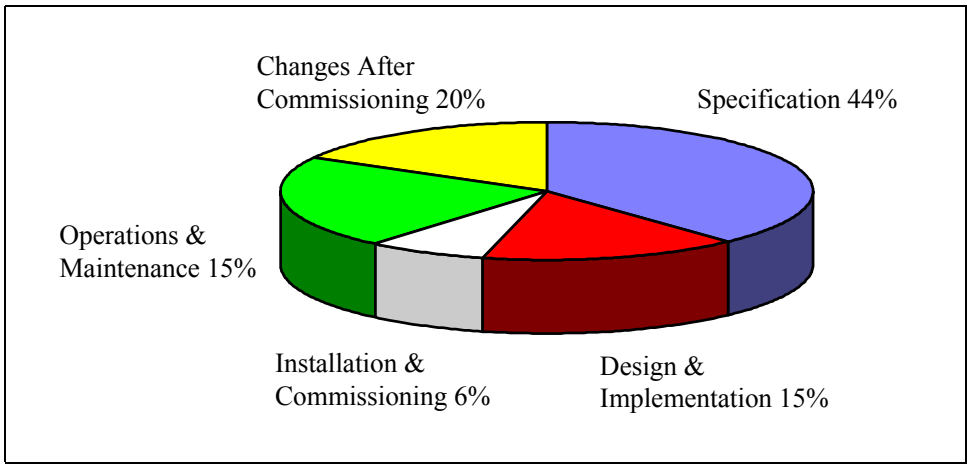


Figure 15-1: Primary Accident Cause by Design Phase, as Published by the U.K. HSE

Overview

This checklist covers the following steps:

- Management Requirements
- Safety Requirements Specification (SRS)
- Conceptual SIS Design
- Detailed SIS Design
- Power & Grounding
- Field Devices
- Operator Interface
- Maintenance/Engineering Interface
- Communications
- Hardware Specification
- Hardware Manufacture
- Application Logic Requirements
- Embedded (Vendor) Software
- Software Coding
- Factory Test
- Installation & Commissioning
- Operations & Maintenance
- Testing
- Management of Change
- Decommissioning

Section 1: Management Requirements

Item #	Item	Circle a choice			Comments
1.1	Have persons or departments responsible for carrying out the phases of the lifecycle been identified?	Y	N	N/A	
1.2	Have persons or departments responsible for carrying out the phases of the lifecycle been informed of their responsibilities?	Y	N	N/A	
1.3	Are persons competent to perform the tasks assigned to them?	Y	N	N/A	
1.4	Is personnel competency documented in terms of knowledge, experience, and training?	Y	N	N/A	
1.5	Has a hazard/risk assessment been performed?	Y	N	N/A	
1.6	Is a safety plan in place that defines the required activities?	Y	N	N/A	
1.7	Are procedures in place to ensure prompt and satisfactory resolution of recommendations?	Y	N	N/A	
1.8	Are procedures in place to audit compliance with the requirements?	Y	N	N/A	

Section 2: Safety Requirements Specification

Item #	Item	Circle a choice			Comments
2.1	Do the safety requirements originate from a systematic hazard assessment? If not, what are the requirements based on?	Y	N	N/A	
2.2	Is there a clear and concise description of each safety related function to be implemented in the SIS?	Y	N	N/A	
2.3	Has the safe state of the process been defined for each operating state of the plant? (e.g., startup, normal operation, maintenance, etc.)	Y	N	N/A	
2.4	Are safety functions defined for each operating state of the plant?	Y	N	N/A	
2.5	Are performance requirements (e.g., speed, accuracy, etc.) defined for each safety function?	Y	N	N/A	
2.6	Has the required safety integrity level (SIL) been determined for each safety function?	Y	N	N/A	
2.7	Are sensor inputs defined with regard to range, accuracy, noise limits, bandwidth, etc.?	Y	N	N/A	
2.8	Are outputs defined with regard to range, accuracy, update frequency, etc.?	Y	N	N/A	
2.9	In the event of system failure, are sufficient information and means available for the operator to assume safe control?	Y	N	N/A	
2.10	Is the operator interface defined in terms of data display, alarms, etc.?	Y	N	N/A	
2.11	Have local or application specific regulatory requirements been considered?	Y	N	N/A	
2.12	Has the operation and implementation of resets been defined for each input and output?	Y	N	N/A	
2.13	Have the operation of bypasses/ overrides been defined for each input and output?	Y	N	N/A	
2.14	Have process common cause considerations (e.g., corrosion, plugging, coating, etc.) been considered?	Y	N	N/A	

Section 3: Conceptual SIS Design

Item #	Item	Circle a choice			Comments
3.1	Are safety functions being handled by a completely separate system from the process control? If not, what is the justification?	Y	N	N/A	
3.2	If multiple functions are being performed within the same logic solver, do the shared components meet the highest SIL requirement?	Y	N	N/A	
3.3	Has the technology and level of redundancy been selected for each safety function? If so, what is it?	Y	N	N/A	
3.4	Have manual test intervals been determined and justified for each safety function?	Y	N	N/A	
3.5	Has the performance of each safety function been analyzed and documented in a quantitative manner in order to see if it meets the safety integrity level (SIL)? If not, what is the justification for the system configuration?	Y	N	N/A	
3.6	Are proven-in-use criteria established for non-certified equipment?	Y	N	N/A	

Section 4: Detailed SIS Design

Item #	Item	Circle a choice			Comments
4.1	Are design documents under control of a formal revision and release program?	Y	N	N/A	
4.2	Has the SIL of the final system been analyzed and documented in a quantitative manner? If not, what is the justification for the system configuration?	Y	N	N/A	
4.3	Are suitable interfaces between field devices and the logic solver defined?	Y	N	N/A	
4.4	Are suitable communication interfaces defined in terms of protocols and information to be exchanged?	Y	N	N/A	
4.5	Are there provisions for future expansion?	Y	N	N/A	
4.6	Are there provisions for incorporating changes as the design proceeds?	Y	N	N/A	
4.7	Is the system fail-safe in terms of:				
	a) Loss of power?	Y	N	N/A	
	b) Loss of instrument air?	Y	N	N/A	
	c) Field cable faults?	Y	N	N/A	
4.8	Can the action of a non-safety function interrupt or compromise any safety functions?	Y	N	N/A	
4.9	Are the safe states of each system component defined?	Y	N	N/A	
4.10	Has the impact of failure of each component in the system been considered, and the required action to be taken defined?	Y	N	N/A	
4.11	Is field I/O power separate from other circuits?	Y	N	N/A	
4.12	Are I/O bypasses incorporated?	Y	N	N/A	
4.13	When an input bypass is enabled, can the state of the sensor still be determined?	Y	N	N/A	
4.14	Are there means for alarming a bypass after a pre-determined time interval?	Y	N	N/A	
4.15	Does the system incorporate manual resetting to restart production? If not, what is the justification?	Y	N	N/A	

Section 5: Power & Grounding

Item #	Item	Circle a choice			Comments
5.1	Are power supplies direct current (DC)? If not, what is the justification?	Y	N	N/A	
5.2	Is a redundant main power source available? If not, what is the justification?	Y	N	N/A	
5.3	Has the impact of power failures been considered?	Y	N	N/A	
5.4	Have the following power concerns been addressed?				
	a) Voltage and current range, including in-rush current?	Y	N	N/A	
	b) Frequency range?	Y	N	N/A	
	c) Harmonics?	Y	N	N/A	
	d) Non-linear loads?	Y	N	N/A	
	e) AC transfer time?	Y	N	N/A	
	f) Overload and short circuit protection?	Y	N	N/A	
	g) Lightning protection?	Y	N	N/A	
	h) Protection against transient spikes, surges, brownouts, and noise?	Y	N	N/A	
	i) Under and over voltage?	Y	N	N/A	
5.5	Have the following grounding concerns been addressed?				
	a) Corrosion protection?	Y	N	N/A	
	b) Cathodic protection?	Y	N	N/A	
	c) Static electricity protection?	Y	N	N/A	
	d) Shield ground?	Y	N	N/A	
	e) Test ground?	Y	N	N/A	
	f) Intrinsic safety barrier grounds?	Y	N	N/A	
	g) Appropriate isolated communications techniques (e.g., communication transformers, fiber optics) between ground planes?	Y	N	N/A	

Section 6: Field Devices

Item #	Item	Circle a choice			Comments
6.1	Is there valid failure rate, failure mode, and diagnostic coverage information for all devices?	Y	N	N/A	
6.2	Have vendors provided recommended manual test intervals and procedures?	Y	N	N/A	
6.3	Will means be available to periodically check the devices for hidden dangerous failures?	Y	N	N/A	
6.4	Are circuits normally energized? If not, is line monitoring (supervised circuits) being incorporated?	Y	N	N/A	
6.5	Does each device have its own dedicated wiring? If not, what is the justification?	Y	N	N/A	
6.6	If smart sensors are being used, are they write-protected?	Y	N	N/A	
6.7	Have minimum, as well as maximum, electrical loads been considered for field I/O circuits?	Y	N	N/A	
6.8	Is feedback available to tell if the final element has moved to its commanded state?	Y	N	N/A	
6.9	Have materials (seals, etc.) been properly selected for the particular application?	Y	N	N/A	
6.10	Does the user have good field experience with the devices in other applications?	Y	N	N/A	
6.11	Are solenoid valves protected from plugging, dirt, insects, freezing, etc.? What measures have been applied?	Y	N	N/A	
6.12	Have the following areas been considered for final elements:				
	a) Opening and closing speeds?	Y	N	N/A	
	b) Shutoff differential pressure?	Y	N	N/A	
	c) Leakage?	Y	N	N/A	
	d) Fire resistance of body, actuator, and impulse lines?	Y	N	N/A	
6.13	Are safety critical field devices identified in some unique manner (e.g., color coding, labeling)?	Y	N	N/A	

Section 7: Operator Interface

Item #	Item	Circle a choice			Comments
7.1	Has failure (loss) of the interface been considered?	Y	N	N/A	
7.2	Are alternate means available to bring the process to a safe state?	Y	N	N/A	
7.3	Are the following shown on the interface:				
	a) Where the process is in a sequence?	Y	N	N/A	
	b) Indication that a SIS action has occurred?	Y	N	N/A	
	c) Indication that a SIS function is bypassed?	Y	N	N/A	
	d) Indication that a SIS component or subsystem has failed or is in a degraded state?	Y	N	N/A	
	e) Status of field devices?	Y	N	N/A	
7.4	Is the update time appropriate for the application under emergency conditions?	Y	N	N/A	
7.5	Have operators been checked for color blindness?	Y	N	N/A	
7.6	Is it possible to change SIS program logic from the operator interface?	Y	N	N/A	
7.7	Do parameters that can be changed have security access protection?	Y	N	N/A	

Section 8: Maintenance/Engineering Interface

Item #	Item	Circle a choice			Comments
8.1	Can failure of this interface adversely affect the SIS?	Y	N	N/A	
8.2	Is there adequate access security? What methods are utilized?	Y	N	N/A	
8.3	Is the maintenance/engineering interface used as the operator interface?	Y	N	N/A	
8.4	Is the maintenance/engineering interface disconnected during normal system operation?	Y	N	N/A	

Section 9: Communications

Item #	Item	Circle a choice			Comments
9.1	Can communication failures have an adverse affect on the SIS?	Y	N	N/A	
9.2	Are communication signals isolated from other energy sources?	Y	N	N/A	
9.3	Has write protection been implemented so that external systems cannot corrupt SIS memory? If not, why?	Y	N	N/A	
9.4	Are interfaces robust enough to withstand EMI/RFI and power disturbances?	Y	N	N/A	

Section 10: Hardware Specification

Item #	Item	Circle a choice			Comments
10.1	Has the physical operating environment been defined and have suitable specifications been set for:				
	a) Temperature range?	Y	N	N/A	
	b) Humidity?	Y	N	N/A	
	c) Vibration and shock?	Y	N	N/A	
	d) Ingress of dust and/or water?	Y	N	N/A	
	e) Contaminating gases?	Y	N	N/A	
	f) Hazardous atmospheres?	Y	N	N/A	
	g) Power supply voltage tolerance?	Y	N	N/A	
	h) Power supply interruptions?	Y	N	N/A	
	i) Electrical interference?	Y	N	N/A	
	j) Ionizing radiation?	Y	N	N/A	
10.2	Are failure modes known for all components?	Y	N	N/A	
10.3	Has the vendor supplied quantitative safe and dangerous system failure rates, including assumptions and component data used?	Y	N	N/A	
10.4	Has the vendor provided diagnostic coverage values for their system?	Y	N	N/A	
10.5	Are logic system components (I/O modules, CPU, communication modules, etc.) all from the same vendor?	Y	N	N/A	
10.6	Has the resulting action of restoring power to the system been considered?	Y	N	N/A	
10.7	Are all I/O modules protected from voltage spikes?	Y	N	N/A	
10.8	If redundant devices or systems are being considered, have measures been taken to minimize potential common cause problems. If so, what are they?	Y	N	N/A	

Section 11: Hardware Manufacture

Item #	Item	Circle a choice			Comments
11.1	Can the vendor provide evidence of an independent safety assessment of the hardware?	Y	N	N/A	
11.2	Does the vendor maintain a formal revision and release control program?	Y	N	N/A	
11.3	Are there visible indications of version numbers on the hardware?	Y	N	N/A	
11.4	Does the vendor have specifications and procedures for the quality of materials, workmanship, and inspections?	Y	N	N/A	
11.5	Are adequate precautions taken to prevent damage due to static discharge?	Y	N	N/A	

Section 12: Application Logic Requirements

Item #	Item	Circle a choice			Comments
12.1	Do all parties have a formal revision and release control program for application logic?	Y	N	N/A	
12.2	Is the logic written in a clear and unambiguous manner that is understandable to all parties?	Y	N	N/A	
12.3	Does the program include comments?	Y	N	N/A	
12.4	Within the logic specification, is there a clear and concise statement of:				
	a) Each safety-related function?	Y	N	N/A	
	b) Information to be given to the operator?	Y	N	N/A	
	c) The required action of each operator command, including illegal or unexpected commands?	Y	N	N/A	
	d) The communication requirements between the SIS and other equipment?	Y	N	N/A	
	e) The initial states for all internal variables and external interfaces?	Y	N	N/A	
	f) The required action for out-of-range variables?	Y	N	N/A	

Section 13: Embedded (Vendor) Software

Item #	Item	Circle a choice			Comments
13.1	Can the vendor provide evidence of an independent safety assessment of all embedded software?	Y	N	N/A	
13.2	Has the software been used in similar applications for a significant period of time?	Y	N	N/A	
13.3	Is the vendor software documented sufficiently for the user to understand its operation and how to implement the desired functionality?	Y	N	N/A	
13.4	Are the results of abnormal math operations fully documented?	Y	N	N/A	
13.5	Are there procedures for the control of software versions in use and the update of all similar systems?	Y	N	N/A	
13.6	For spares which contain firmware, is there a procedure to ensure all modules are compatible?	Y	N	N/A	
13.7	Can software versions in use easily be checked?	Y	N	N/A	
13.8	If errors are found in embedded software, are they reported to and corrected by the vendor, and incorporated into the SIS only after checking and testing the corrected code?	Y	N	N/A	
13.9	Does the manufacturer provide competent technical support?	Y	N	N/A	

Section 14: Software Coding

Item #	Item	Circle a choice			Comments
14.1	Are there standards and procedures for software coding?	Y	N	N/A	
14.2	Are there procedures for documenting and correcting any deficiencies in the specification or design revealed during the coding phase?	Y	N	N/A	
14.3	Are departures from or enhancements to the requirements of the design documented?	Y	N	N/A	
14.4	Is a formal language or some other means taken to assure the program is both precise and unambiguous?	Y	N	N/A	
14.5	Is there a procedure for generating and maintaining adequate documentation?	Y	N	N/A	
14.6	Does the programming language encourage the use of small and manageable modules?	Y	N	N/A	
14.7	Does the code include adequate comments?	Y	N	N/A	
14.8	Are design reviews carried out during program development involving users, designers, and programmers?	Y	N	N/A	
14.9	Does the software contain adequate error detection facilities associated with error containment, recovery, or safe shutdown?	Y	N	N/A	
14.10	Are all functions testable?	Y	N	N/A	
14.11	Is the final code checked against the requirements by persons other than those producing the code?	Y	N	N/A	
14.12	Is a well-established compiler/assembler used?	Y	N	N/A	
14.13	Is the compiler/assembler certified to recognized standards?	Y	N	N/A	

Section 15: Factory Test

Item #	Item	Circle a choice			Comments
15.1	Are there procedures for testing the finished system?	Y	N	N/A	
15.2	Are records maintained of test results?	Y	N	N/A	
15.3	Are there procedures for documenting and correcting any deficiencies in the specification, design or programming revealed during testing?	Y	N	N/A	
15.4	Is testing carried out by persons other than those producing the code?	Y	N	N/A	
15.5	Is software tested in the target system rather than simulated?	Y	N	N/A	
15.6	Is each control flow or logic path tested?	Y	N	N/A	
15.7	Have arithmetic functions been tested with minimum and maximum values to ensure that no overflow conditions are reached?	Y	N	N/A	
15.8	Are there tests to simulate exceptions as well as normal conditions?	Y	N	N/A	
15.9	Have all of the following items been tested?	Y	N	N/A	
	a) Dependence on other systems/ interfaces	Y	N	N/A	
	b) Logic solver configuration	Y	N	N/A	
	c) Operation of bypasses	Y	N	N/A	
	d) Operation of resets	Y	N	N/A	
	e) All functional logic	Y	N	N/A	

Section 16: Installation & Commissioning

Item #	Item	Circle a choice			Comments
16.1	Have personnel received appropriate training?	Y	N	N/A	
16.2	Is there sufficient independence between those carrying out the work and those inspecting it?	Y	N	N/A	
16.3	Have adequate precautions been taken for storage of items during installation?	Y	N	N/A	
16.4	Are installation procedures for all devices sufficient in detail so as not to leave important interpretations or decisions to installation personnel?	Y	N	N/A	
16.5	Has the SIS been inspected in order to reveal any damage caused during installation?	Y	N	N/A	
16.6	Are items such as cabinets, junction boxes, and cables protected from:				
	a) Steam leaks?	Y	N	N/A	
	b) Water leaks?	Y	N	N/A	
	c) Oil leaks?	Y	N	N/A	
	d) Heat sources?	Y	N	N/A	
	e) Mechanical damage?	Y	N	N/A	
	f) Corrosion (e.g., process fluids flowing from damaged sensors to junction boxes, the logic cabinet, or the control room)?	Y	N	N/A	
	g) Combustible atmospheres?	Y	N	N/A	
16.7	Are safety related systems clearly identified to prevent inadvertent tampering?	Y	N	N/A	
16.8	Has the proper operation of the following items been confirmed?				
	a) Proper installation of equipment and wiring?	Y	N	N/A	
	b) Energy sources are operational?	Y	N	N/A	
	c) All field devices have been calibrated?	Y	N	N/A	
	d) All field devices are operational?	Y	N	N/A	
	e) Logic solver is operational?	Y	N	N/A	
	f) Communication with other systems?	Y	N	N/A	
	g) Operation and indication of bypasses?	Y	N	N/A	
	h) Operation of resets?	Y	N	N/A	
	i) Operation of manual shutdowns?	Y	N	N/A	

16.9	Is the documentation consistent with the actual installation?	Y	N	N/A	
16.10	Is there documentation showing the following:				
	a) Identification of the system being commissioned?	Y	N	N/A	
	b) Confirmation that commissioning has been successfully completed?	Y	N	N/A	
	c) The date the system was commissioned?	Y	N	N/A	
	d) The procedures used to commission the system?	Y	N	N/A	
	e) Authorized signatures indicating the system was successfully commissioned?	Y	N	N/A	

Section 17: Operations & Maintenance

Item #	Item	Circle a choice			Comments
17.1	Have employees been adequately trained on the operating and maintenance procedures for the system?	Y	N	N/A	
17.2	Are operating procedures adequately documented?	Y	N	N/A	
17.3	Is there a user/operator/maintenance manual for the system?	Y	N	N/A	
17.4	Does the manual describe:				
	a) Limits of safe operation, and the implications of exceeding them?	Y	N	N/A	
	b) How the system takes the process to a safe state?	Y	N	N/A	
	c) The risk associated with system failures and the actions required for different failures?	Y	N	N/A	
17.5	Are there means to limit access only to authorized personnel?	Y	N	N/A	
17.6	Can all operational settings be readily inspected to ensure they are correct at all times?	Y	N	N/A	
17.7	Are there means to limit the range of input trip settings?	Y	N	N/A	
17.8	Have adequate means been established for bypassing safety functions?	Y	N	N/A	
17.9	When functions are bypassed, are they clearly indicated?	Y	N	N/A	
17.10	Have documented procedures been established to control the application and removal of bypasses?	Y	N	N/A	
17.11	Have documented procedures been established to ensure the safety of the plant during SIS maintenance?	Y	N	N/A	
17.12	Are maintenance procedures sufficient in detail so as not to leave important interpretations or decisions to maintenance personnel?	Y	N	N/A	
17.13	Are maintenance activities and schedules defined for all portions of the system?	Y	N	N/A	
17.14	Are procedures periodically reviewed?	Y	N	N/A	
17.15	Are procedures in place to prevent unauthorized tampering?	Y	N	N/A	

17.16	Are there means to verify that repair is carried out in a time consistent with that assumed in the safety assessment?	Y	N	N/A	
17.17	Are maintenance and operational procedures in place to minimize the introduction of potential common cause problems?	Y	N	N/A	
17.18	Is the documentation consistent with the actual maintenance and operating procedures?	Y	N	N/A	

Section 18: Testing

Item #	Item	Circle a choice			Comments
18.1	Are documented provisions and procedures in place to allow proof testing of all safety functions, including field devices?	Y	N	N/A	
18.2	Are test procedures sufficient in detail so as not to leave important interpretations or decisions to maintenance personnel?	Y	N	N/A	
18.3	Has the basis for the periodic test interval been documented?	Y	N	N/A	
18.4	Are the following items tested:				
	a) Impulse lines?	Y	N	N/A	
	b) Sensing devices?	Y	N	N/A	
	c) Application logic, computations, and/or sequences?	Y	N	N/A	
	d) Trip points?	Y	N	N/A	
	e) Alarm functions?	Y	N	N/A	
	f) Speed of response?	Y	N	N/A	
	g) Final elements?	Y	N	N/A	
	h) Manual trips?	Y	N	N/A	
	i) Diagnostics?	Y	N	N/A	
18.5	Is there a fault reporting system?	Y	N	N/A	
18.6	Are procedures in place to compare actual performance against the predicted or required performance?	Y	N	N/A	
18.7	Are there documented procedures for correcting any deficiencies found?	Y	N	N/A	
18.8	Is calibration of test equipment verified?	Y	N	N/A	
18.9	Are test records maintained?	Y	N	N/A	
18.10	Do test records show:				
	a) Date of inspection/test?	Y	N	N/A	
	b) Name of person conducting test?	Y	N	N/A	
	c) Identification of device being tested?	Y	N	N/A	
	d) Results of test?	Y	N	N/A	
18.11	Are testing procedures in place to minimize the introduction of potential common cause problems?	Y	N	N/A	
18.12	Is failure rate data periodically reviewed and compared with data used during system design/analysis?	Y	N	N/A	

Section 19: Management of Change

Item #	Item	Circle a choice			Comments
19.1	Are there approval procedures which consider the safety implications of all modifications, such as:				
	a) Technical basis for the change?	Y	N	N/A	
	b) Impact on safety and health?	Y	N	N/A	
	c) Impact on operating/maintenance procedures?	Y	N	N/A	
	d) Time required?	Y	N	N/A	
	e) Effect on response time?	Y	N	N/A	
19.2	Are there procedures that define the level of review/approval required depending upon the nature of the change?	Y	N	N/A	
19.3	Has the proposed change initiated a return to the appropriate phase of the lifecycle?	Y	N	N/A	
19.4	Has the project documentation (e.g., operating, test, maintenance procedures, etc.) been altered to reflect the change?	Y	N	N/A	
19.5	Has the complete system been tested after changes have been introduced, and the results documented?	Y	N	N/A	
19.6	Are there documented procedures to verify that changes have been satisfactorily completed?	Y	N	N/A	
19.7	Have all affected departments been appraised of the change?	Y	N	N/A	
19.8	Is access to the hardware and software limited to authorized and competent personnel?	Y	N	N/A	
19.9	Is access to the project documentation limited to authorized personnel?	Y	N	N/A	
19.10	Are project documents subject to appropriate revision control?	Y	N	N/A	
19.11	Have the consequences of incorporating new versions of software been considered?	Y	N	N/A	

Section 20: Decommissioning

Item #	Item	Circle a choice			Comments
20.1	Have management of change procedures been followed for decommissioning activities?	Y	N	N/A	
20.2	Has the impact on adjacent operating units and facilities been evaluated?	Y	N	N/A	
20.3	Are there procedures to maintain the safety of the process during decommissioning?	Y	N	N/A	
20.4	Are there procedures that define the level of authorization required for decommissioning?	Y	N	N/A	

References

1. *Programmable Electronic Systems in Safety Related Applications - Part 2 - General Technical Guidelines.* U.K. Health & Safety Executive, 1987.
2. ISA-84.01-1996. *Application of Safety Instrumented Systems for the Process Industries*.
3. *Guidelines for Safe Automation of Chemical Processes*. American Institute of Chemical Engineers - Center for Chemical Process Safety, 1993.
4. *Out of Control: Why control systems go wrong and how to prevent failure.* U.K. Health & Safety Executive, 1995.